

A Split Resolution of the Functional Zauner Conjectures: Degenerate Existence and a p -Adic Impossibility

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Source targets and result

The source is K. Mahesh Krishna's *Non-Archimedean and p -adic Functional Welch Bounds*. It is arXiv:2209.06769 and formulates a non-Archimedean functional Zauner conjecture (Conjecture 2.5), a p -adic functional Zauner conjecture (Conjecture 3.4), two equality questions, and two functional equiangular-line problems.

This packet proves four exact conclusions.

- (1) Under the source field axiom, every positive integer has absolute value one. On $\mathbb{K}^d$ with the standard maximum norm, a repeated vector–functional pair solves Question 2.4 for every d and $n \geq 2$, and proves Conjecture 2.5 as literally stated.
- (2) Conjecture 3.4 is false: for every prime p , no proposed family can exist in any dimension d divisible by p .
- (3) In dimension one, Question 3.3 has a complete answer for $n \geq 2$: such a family exists exactly when $p \nmid n$.
- (4) In both functional equiangular formulations, the parameter choice $(a, \gamma) = (1, 1)$ admits arbitrarily many repetitions. Hence the stated problems have no finite Gerzon bound without a distinctness hypothesis.

The decisive source conjectures are reproduced below.

spaces, [24] for Zauner conjecture in non-Archimedean Hilbert spaces and [22] for Zauner conjecture in p -adic Hilbert spaces).

Conjecture 2.5. (Non-Archimedean Functional Zauner Conjecture) *Let $\mathbb{K}$ non-Archimedean field satisfying Equation (3). For each $d \in \mathbb{N}$, there exist vectors $\tau_1, \dots, \tau_{d^2} \in \mathbb{K}^d$ (w.r.t. any non-Archimedean norm) and functionals $f_1, \dots, f_{d^2} \in (\mathbb{K}^d)^*$ satisfying the following.*

- (i) $f_j(\tau_j) = 1$ for all $1 \leq j \leq d^2$.
- (ii) The operator $S_{f, \tau} : \mathbb{K}^d \ni x \mapsto \sum_{j=1}^{d^2} f_j(x) \tau_j \in \mathbb{K}^d$ is diagonalizable.
- (iii)

$$|f_j(\tau_k) f_k(\tau_j)| = |d|, \quad \forall 1 \leq j, k \leq d^2, j \neq k.$$

- (iv) $\|f_j\| = 1$ for all $1 \leq j \leq d^2$, $\|\tau_j\| = 1$ for all $1 \leq j \leq d^2$.

There are four bounds which are companions of Welch bounds in Hilbert spaces. To recall them we need the notion of Gerzon's bound.

Definition 2.6. [45] *Given $d \in \mathbb{N}$, define Gerzon's bound*

$$\mathcal{Z}(d, \mathbb{K}) := \begin{cases} d^2 & \text{if } \mathbb{K} = \mathbb{C} \\ \underline{d(d+1)} & \text{if } \mathbb{K} = \mathbb{R} \end{cases}$$

Conjecture 3.4. (*p*-adic Functional Zauner Conjecture) *Let p be a prime. For each $d \in \mathbb{N}$, there exist vectors $\tau_1, \dots, \tau_{d^2} \in \mathbb{Q}_p^d$ (w.r.t. any non-Archimedean norm) and functionals $f_1, \dots, f_n \in (\mathbb{Q}_p^d)^*$ satisfying the following.*

(i) $f_j(\tau_j) = 1$ for all $1 \leq j \leq d^2$.

(ii) There exists $b \in \mathbb{Q}_p$ satisfying

$$\sum_{j=1}^{d^2} f_j(x) \tau_j = bx, \quad \forall x \in \mathcal{X},$$

(iii)

$$|f_j(\tau_k) f_k(\tau_j)| = |d|, \quad \forall 1 \leq j, k \leq d^2, j \neq k.$$

(iv) $\|f_j\| = 1$ for all $1 \leq j \leq d^2$, $\|\tau_j\| = 1$ for all $1 \leq j \leq d^2$.

Theorem 2.7 and Theorem 3.1 give the following problem.

Question 3.5. *Whether there is a p -adic functional version of Theorem 2.7? In particular, does there exist a version of*

(i) *p -adic functional Bukh-Cox bound?*

(ii) *p -adic functional Orthoplex/Rankin bound?*

(iii) *p -adic functional Levenstein bound?*

(iv) *p -adic functional Exponential bound?*

We end by formulating p -adic functional equiangular line problem.

The non-Archimedean functional half

The source assumes

$$\left| \sum_{j=1}^r \lambda_j^2 \right| = \max_{1 \leq j \leq r} |\lambda_j|^2 \quad \text{for every } r \geq 1 \text{ and } \lambda_j \in \mathbb{K}. \quad (1)$$

Lemma 1. *If $\mathbb{K}$ satisfies (1), then $|r| = 1$ for every positive integer r . In particular, $\mathbb{K}$ has characteristic zero.*

Proof. Take $\lambda_1 = \dots = \lambda_r = 1$ in (1). □

Theorem 1. *Let $\mathbb{K}$ satisfy (1). Equip $\mathbb{K}^d$ with the maximum norm. For every $d \geq 1$ and $n \geq 2$, all conditions of Question 2.4 hold. Taking $n = d^2$ proves Conjecture 2.5 under the same literal reading. More generally, the construction works in any normed space containing a norming pair (u, φ) with $\|u\| = \|\varphi\| = \varphi(u) = 1$.*

Proof. For the maximum norm take $u = e_1$ and $\varphi(x_1, \dots, x_d) = x_1$, and repeat

$$\tau_j = u, \quad f_j = \varphi \quad (1 \leq j \leq n).$$

Then $f_j(\tau_j) = 1$, both norms are one, and

$$S_{f, \tau} x = \sum_{j=1}^n \varphi(x) u = n \varphi(x) u.$$

Since $P = u \otimes \varphi$ satisfies $P^2 = P$, the operator nP is diagonalizable. For $j \neq k$, $|f_j(\tau_k) f_k(\tau_j)| = 1$. Lemma 1 gives $|n| = |d| = 1$, so Question 2.4's equality reads

$$\max\{1, 1\} = 1 = \frac{|n|^2}{|d|}.$$

For $n = d^2$, Conjecture 2.5's off-diagonal requirement is $1 = |d|$, which also holds. $\square$

The phrase “with respect to any non-Archimedean norm” is ambiguous. The theorem proves existence for the standard norm and for every norm admitting a norming pair. It does not invoke a non-Archimedean Hahn–Banach theorem over an arbitrary nonspherically-complete field.

The p -adic conjecture is impossible when $p \mid d$

We read the source's undefined terminal n in Conjecture 3.4 as d^2 , as required by the displayed vectors, the sum, and conditions (i), (iii), and (iv). The contradiction below is independent of all norm choices.

Theorem 2. *Fix a prime p . If $p \mid d$, no vectors and functionals can satisfy Conjecture 3.4 in $\mathbb{Q}_p^d$. Consequently the universal p -adic functional Zauner conjecture is false.*

Proof. Suppose the conjectural family existed, and write

$$Sx = \sum_{k=1}^{d^2} f_k(x) \tau_k = bx.$$

The trace of the rank-one operator $\tau_k \otimes f_k$ is $f_k(\tau_k) = 1$. Taking traces therefore gives

$$bd = \operatorname{tr} S = \sum_{k=1}^{d^2} f_k(\tau_k) = d^2, \quad \text{hence } b = d.$$

Fix j . Applying f_j to $S\tau_j = d\tau_j$ yields

$$d = 1 + \sum_{k \neq j} f_k(\tau_j) f_j(\tau_k). \quad (2)$$

Every summand on the right after the initial 1 has p -adic absolute value $|d|_p$, by condition (iii). If $p \mid d$, their sum has absolute value at most $|d|_p < 1$. The ultrametric inequality then says that the right side of (2) has absolute value exactly 1, while its left side has absolute value $|d|_p < 1$, a contradiction. $\square$

This also exposes a mismatch in the source's claim that Conjecture 3.4 is a special case of Question 3.3. When $n = d^2$ and $p \mid d$, the conjecture sets every off-diagonal product modulus to $|d|_p$, whereas the equality in Question 3.3 has right side $|d^2|_p^2 / |d|_p = |d|_p^3$; these cannot agree.

Exact one-dimensional p -adic classification

Theorem 3. *For $d = 1$, $n \geq 2$, and the standard norm on $\mathbb{Q}_p$, Question 3.3 has a solution if and only if $p \nmid n$.*

Proof. In one dimension write $\tau_j = a_j$ and $f_j(x) = c_j x$. The condition $f_j(\tau_j) = 1$ gives $c_j a_j = 1$, and hence, for every $j \neq k$,

$$f_j(\tau_k) f_k(\tau_j) = (c_j a_k)(c_k a_j) = 1.$$

Thus the left side of condition (iii) is 1, while its right side is $|n|_p^2$. Equality is possible exactly when $|n|_p = 1$, or $p \nmid n$. Conversely, if $p \nmid n$, take every $\tau_j = 1$ and every f_j equal to the identity functional. Then $Sx = nx$, and all four conditions hold. $\square$

No finite functional Gerzon bound as stated

Corollary 1. *For every $d \geq 1$, both $n(\mathbb{K}, 1, d, 1)$ and $n(p, 1, d, 1)$ are unbounded under the source’s displayed definitions (using the standard maximum norm).*

Proof. For every prescribed N , repeat N copies of e_1 and its first coordinate functional. Every self-pairing is one, every cross-product is one in absolute value, and all norms are one. The source does not require the vectors, functionals, or represented lines to be distinct. $\square$

Scope, duplicate boundary, and verification

This is a full counterexample to the universal p -adic functional Zauner conjecture, a full solution of the one-dimensional equality question, and a literal solution of the non-Archimedean statement for the standard norm. It does not construct the four Bukh–Cox/Rankin/Levenstein/exponential analogues, classify Question 3.3 in higher dimensions, or solve a strengthened distinct-line problem.

The run contains adjacent packets for arXiv:2210.07062 (the vector, rather than functional, repeated-vector degeneracy) and arXiv:2209.12833 (a continuous diagonal-measure obstruction). Neither contains the trace and diagonal identity (2), the $p \mid d$ impossibility, or the one-dimensional functional classification proved here. Exact-id and core keyword searches found no duplicate. Bounded primary-source web searches found only the source and its continuous companion, with no later correction or resolution.

Human review should focus on the natural $n = d^2$ repair of the source typo, the trace identity, and the intended force of “any norm.” The p -adic counterexample itself is norm-independent.

References

- [1] K. Mahesh Krishna, *Non-Archimedean and p -adic Functional Welch Bounds*, arXiv:2209.06769 (2022).